

Jerry (Jiarui) Hu

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EDUCATION

Sept 2026 – Expected 2030 | University of Illinois Urbana-Champaign (Grainger College of Engineering)

B.S. in Computer Engineering

Prev: Sept 2021 – May 2026 | White Rock Christian Academy (WRCA) — International Baccalaureate Candidate

SKILLS

Languages: Java, C, Python, Swift, SQL, HTML/CSS/JavaScript, C++

Embedded & Hardware: ESP32/MicroPython, PCB Design + Soldering, MQTT, 3D CAD, 3D Printing

Frameworks & Tools: Node.JS, React, Google Firebase, Tailwind, Figma, KiCAD, Autodesk Fusion360

EXPERIENCE

Sept 2023 – June 2026 | Constellation Networking (constellationhub.org)

Chief Technology Officer (Co-founder) — *Cloudflare, Python, Firebase (Firestore, Auth, Storage, Cloud Functions)*

- Designed + built a networking platform for peers, 1.5k unique monthly visitors; recruited + led 3-person dev team
- Utilized Firebase Cloud Functions for profile matching between students and professionals

Feb 2024 – June 2026 | Millennium STEM BC

Director of IT — *GitHub Pages, Firebase (Storage, Cloud Functions), TailwindCSS, HTML/JavaScript*

- Identified severe vulnerabilities on organization website; was then recruited as Director of IT and rebuilt website
- Built database containing 70+ STEM volunteer opportunities; hosted annual stem conference of 300+ attendees

Aug 2024 – Feb 2025 | National High School Data Analysis & AI Challenge

Research Team Lead — *Python, NumPy, Scikit-learn, Matplotlib, Plotly*

- Initiated research topic and led 5-person team; analyzed youth substance-use trends across Canada
- Co-authored and published research paper in STEM Fellowship Journal; advanced to STEM Fellowship Big Data & AI Challenge National Finals

PERSONAL PROJECTS

Oct 2024 | WiFi Triangulation Algorithm

Python, NumPy, Scikit-learn, Plotly 3D, Matplotlib

- Wrote a Python library to extract RSSI (proximity) data from a mesh router network
- Trained Random Forest Model on labeled samples to predict a device's location with room-level-accuracy

July – Aug 2025 | Smart Home Garage Door Controller

ESP32 BLE Beacon + Self-designed Relay PCB, KiCAD, Fusion360

- Designed a relay PCB to interface with garage door motor controller; integrated to open source project Home Assistant
- Built a proximity-trigger using BLE RSSI distance estimation with a BLE-beacon in car to actuate the door

Apr – May 2025 | Motion Sensor System

ESP32-C2, PIR Sensor, KiCAD, Fusion360, MQTT, Home Assistant

- Designed custom PCBs and 3D designed + printed enclosures for 10 hand-soldered PIR motion sensors; integrated via MQTT to Home Assistant for real-time occupancy reporting

Sept – Nov 2025 | Custom Airflow Controller for IB Physics Research

ESP32, KiCAD, Power control

- Built a PWM-controlled brushless DC fan system with 10 discrete airflow levels and a servo-driven automated trial mechanism; designed PCB in KiCAD and model in Fusion360

AWARDS & LEADERSHIP

- **Honour Roll, Top 5% in Canada (Canadian Computing Competition)** — Junior (2024)
- **Top 25% in Canada (Canadian Computing Competition)** — Senior (2025)
- **1st place, Best Overall Hack (Major Leagues Hacking)** — Hacks for Hackers Hackathon (2024)
- **2nd place, Solving Global Challenges (Yale Young Global Scholars)** — co-designed low-cost water pollution monitor bot (2024)
- **Founder, Computing Club** (grew to 21 members) | **Concertmaster, WRCA Orchestra** (3 years) | **MVP, WRCA Badminton** (2 years)